

MedShield Twin

AI-Powered Smart Hospital Security Audit Report | Generated: 2026-07-31 03:13:27 by System Test Auditor

1. Executive Security Summary

This security report provides a cryptographic and analytical evaluation of the **MedShield Twin** digital hospital infrastructure. The twin currently monitors **16** active Internet of Medical Things (IoMT) devices distributed across clinical departments. Currently, **11** devices are functioning normally, **4** are flagged under active AI cyber-threat warnings, and **1** are powered down or disconnected. Cryptographic verification of historical records reports the system blockchain is in a **WARNING / TAMPERED** state.

Security Parameter	Current Value	Evaluation
Hospital Devices Tracked	16	Baseline Active Layout
Active AI Threat Warnings	4	Requires Immediate Isolation
Blockchain Ledger Audit	WARNING / TAMPERED	SECURITY ALERT: Records Tampered!

2. Departmental IoMT Device Audit

Below is the risk-assessment catalog for smart medical equipment by department:

Device ID	Name	Room / Department	Status	Risk Score
ICU-ECG-01	ECG Monitor	ICU	Online	0%
ICU-VEN-01	Smart Ventilator	ICU	Under Attack	72%
ICU-PMP-01	Infusion Pump	ICU	Offline	0%
ER-MON-01	Patient Monitor	Emergency Room	Online	0%
ER-BPS-01	Blood Pressure Sensor	Emergency Room	Online	0%
ER-OXY-01	Oxygen Sensor	Emergency Room	Online	0%
OT-ANS-01	Anesthesia Machine	Operation Theatre	Online	0%
OT-SYS-01	Surgical Navigation System	Operation Theatre	Online	0%
RAD-MRI-01	MRI Controller	Radiology	Under Attack	0%
RAD-XRY-01	X-Ray Modality	Radiology	Online	0%
PHR-DIS-01	Automated Drug Dispenser	Pharmacy	Under Attack	0%
LAB-ANA-01	Blood Analyzer	Laboratory	Online	0%
GW-BED-01	Smart Bed Sensor	General Ward	Online	0%
SRV-DB-01	Hospital Core Database	Server Room	Under Attack	76%

NET-RTR-01	Main Firewall Router	Network Control Room	Online	0%
REC-KIOSK-01	Check-in Kiosk	Reception	Online	0%

3. AI Threat History logs

The Machine Learning Random Forest classifier flagged the following security anomalies:

Timestamp	Device ID	Severity	Alert Message Description
2026-07-31 03:12:25	SRV-DB-01	Critical	AI Threat Detected: Spoofing on Hospital Core Database. ML Confidence: 100.0% - Packet spoofing detected. The payload anomaly rating is very high (0.85), suggesting injected shellcode or modified patient sensor metrics.
2026-07-31 03:11:59	ICU-VEN-01	Critical	AI Threat Detected: DDoS on Smart Ventilator. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.
2026-07-31 03:11:55	ICU-VEN-01	Critical	AI Threat Detected: DDoS on Smart Ventilator. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.
2026-07-31 03:11:40	ICU-VEN-01	Critical	AI Threat Detected: DDoS on Smart Ventilator. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.
2026-07-31 03:11:23	ER-MON-01	Critical	AI Threat Detected: Spoofing on Patient Monitor. ML Confidence: 100.0% - Packet spoofing detected. The payload anomaly rating is very high (0.85), suggesting injected shellcode or modified patient sensor metrics.
2026-07-31 03:10:14	NET-RTR-01	High	AI Threat Detected: Brute Force on Main Firewall Router. ML Confidence: 92.27% - Brute force attempt detected. There are multiple failed logins (15) observed, typical of password guessing or credential stuffing on medical consoles.
2026-07-31 03:09:04	SRV-DB-01	High	AI Threat Detected: Port Scan on Hospital Core Database. ML Confidence: 100.0% - Port scan detected. The port entropy is high (0.95), showing that a single IP is trying to probe multiple ports in a short duration.
2026-07-31 03:08:53	SRV-DB-01	High	AI Threat Detected: Port Scan on Hospital Core Database. ML Confidence: 100.0% - Port scan detected. The port entropy is high (0.95), showing that a single IP is trying to probe multiple ports in a short duration.

2026-07-31 03:08:47	SRV-DB-01	High	AI Threat Detected: Port Scan on Hospital Core Database. ML Confidence: 100.0% - Port scan detected. The port entropy is high (0.95), showing that a single IP is trying to probe multiple ports in a short duration.
2026-07-31 03:08:22	ICU-VEN-01	Critical	AI Threat Detected: DDoS on Smart Ventilator. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.
2026-07-31 03:08:15	ICU-VEN-01	Critical	AI Threat Detected: DDoS on Smart Ventilator. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.
2026-07-31 03:08:12	ICU-VEN-01	Critical	AI Threat Detected: DDoS on Smart Ventilator. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.
2026-07-31 03:07:29	NET-RTR-01	High	AI Threat Detected: Brute Force on Main Firewall Router. ML Confidence: 92.27% - Brute force attempt detected. There are multiple failed logins (15) observed, typical of password guessing or credential stuffing on medical consoles.
2026-07-31 03:07:23	NET-RTR-01	High	AI Threat Detected: Brute Force on Main Firewall Router. ML Confidence: 92.27% - Brute force attempt detected. There are multiple failed logins (15) observed, typical of password guessing or credential stuffing on medical consoles.
2026-07-31 03:07:16	SRV-DB-01	High	AI Threat Detected: Port Scan on Hospital Core Database. ML Confidence: 100.0% - Port scan detected. The port entropy is high (0.95), showing that a single IP is trying to probe multiple ports in a short duration.
2026-07-31 03:07:06	NET-RTR-01	Critical	AI Threat Detected: DDoS on Main Firewall Router. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.
2026-07-31 03:06:46	NET-RTR-01	Critical	AI Threat Detected: DDoS on Main Firewall Router. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.

2026-07-31 03:06:42	NET-RTR-01	Critical	AI Threat Detected: DDoS on Main Firewall Router. ML Confidence: 88.0% - DDoS flooding detected. Packet rate is extremely high (1500.0 packets/sec), which overloads medical devices.
2026-07-31 03:06:29	GW-BED-01	Critical	AI Threat Detected: Spoofing on Smart Bed Sensor. ML Confidence: 100.0% - Packet spoofing detected. The payload anomaly rating is very high (0.85), suggesting injected shellcode or modified patient sensor metrics.
2026-07-31 03:06:22	GW-BED-01	Critical	AI Threat Detected: Spoofing on Smart Bed Sensor. ML Confidence: 100.0% - Packet spoofing detected. The payload anomaly rating is very high (0.85), suggesting injected shellcode or modified patient sensor metrics.

4. Recommended Mitigation Strategies

1. Departmental Network Segmentation: Isolate clinical IoMT devices (ICU, Operation Theatre) from standard hospital WiFi networks. Apply firewalls restricting inter-departmental traffic.

2. Machine Learning Pipeline updates: Regularly update baseline dataset inputs in the *train.py* pipeline to prevent ML classification drift on newer Botnet signatures.

3. Cryptographic Ledger Backup: In case of 'TAMPERED' warnings, cross-reference the digital signatures (SIG_Admin, SIG_Analyst) stored inside the database with the secure cold backup logs.

4. Firmware Patching: Instantly isolate and patch firmware on devices indicating elevated risk scores (> 40%) to eliminate buffer overflows or protocol exploits.